



Title

rdwinselect — Window selection procedure for RD designs under local randomization.

Syntax

```
rdwinselect runvar [covariates] [if] [in] [, cutoff(#) obsmin(#) wmin(# #) wobs(#)
wstep(#) wasymmetric wmasspoints nwindows(#) dropmissing statistic(stat_name)
p(#) evalat(point) kernel(kerneltype) approximate level(#) reps(#) seed(#)
plot graph_options(graphopts) genvars obsstep(#) ]
```

Description

rdwinselect implements a window selection procedure based on balance tests for regression discontinuity (RD) designs under local randomization. Specifically, it constructs a sequence of nested windows around the RD cutoff and reports binomial tests for the running variable *runvar* and covariate balance tests for covariates *covariates* (if specified). The recommended window is the largest window around the cutoff such that the minimum p-value from the balance tests is larger than a prespecified level for all nested (smaller) windows. By default, the p-values are calculated using randomization inference methods. See [Cattaneo, Frandsen and Titiunik \(2015\)](#) and [Cattaneo, Titiunik and Vazquez-Bare \(2017\)](#) for an introduction to this methodology.

A detailed introduction to this command is given in [Cattaneo, Titiunik and Vazquez-Bare \(2016\)](#).

Companion R and Python packages are also available [here](#).

Companion functions are [rdrandinf](#), [rdsensitivity](#) and [rdrbounds](#).

Related R, Python, and Stata packages useful for inference in RD designs are described in the following website:

<https://rdpackages.github.io/>

Options

cutoff(#) specifies the RD cutoff for the running variable *runvar*. Default is **cutoff(0)**.

_____ Window selection _____

obsmin(#) specifies the minimum number of observations above and below the cutoff in the smallest window. Default is **obsmin(10)**.

wmin(# #) specifies the initial window to be used (if **obsmin(#)** is not specified). Can be a single number to specify the length of the (symmetric) initial window, or two numbers to specify the left and right limits of the initial window. Specifying both **wmin(#)** and **obsmin(#)** returns an error.

wobs(#) specifies the number of observations to be added on each side of the cutoff at each step. Default is **wobs(5)**.

wstep(#) specifies the increment in window length. Specifying both **wobs(#)** and **wstep(#)** returns an error.

wasymmetric allows for asymmetric windows around the cutoff (when **wobs(#)** is specified).

wmasspoints specifies that the running variable is discrete and each masspoint should be used as a window.

nwindows(#) specifies the number of windows to be used. Default is **nwindows(10)**.

dropmissing drops rows with missing values in covariates when calculating windows.

Statistic

statistic(*stat_name*) specifies the statistic to be used. Options are:
diffmeans for difference in means statistic. This is the default option.
ksmirnov for Kolmogorov-Smirnov statistic.
ranksum for Wilcoxon-Mann-Whitney studentized statistic.
hotelling for Hotelling's T-squared statistic.
The option **ttest** is equivalent to **diffmeans** and included for backward compatibility.

p(#) specifies the order of the polynomial for outcome adjustment model. Default is **p(0)**.

evalat(*point*) specifies the point at which the adjusted variable is evaluated. Allowed options are **cutoff** and **means**. Default is **evalat(cutoff)**.

kernel(*kerneltype*) specifies the type of kernel to use as a weighting scheme. Allowed kernel types are **uniform** (uniform kernel), **triangular** (triangular kernel), and **epan** (Epanechnikov kernel). Default is **kernel(uniform)**.

Inference

approximate specifies that covariate balance tests should use a large-sample approximation instead of finite-sample exact randomization inference methods.

level(#) specifies the minimum accepted value of the p-value from the covariate balance tests to be used. Default is **level(.15)**.

reps(#) specifies the number of replications to be used. Default is **reps(1000)**.

seed(#) sets the seed for the randomization test. With this option, the user can manually set the desired seed, or can enter the value -1 to use the system seed. Default is **seed(666)**.

Generate plots and variables

plot draws a scatter plot of the minimum p-value from the covariate balance test against window length implemented by the command.

graph_options(*graphopts*) graph options for plot generated by the command.

genvars generates a variable indicating the window number corresponding to each observation and a variable indicating the corresponding window length.

Backward compatibility

obsstep(#) specifies the minimum number of observations to be added on each side of the cutoff. This option is deprecated and only included for backward compatibility. We recommend the use of **wstep** or **wobs** instead.

Example: Cattaneo, Frandsen and Titiunik (2015) Incumbency Data

Setup

```
. use rdlocrand_senate.dta, clear
```

Window selection with three covariates and default options

```
. rdwinselect demmv dopen population demvoteslag1
```

Window selection using Kolmogorov-Smirnov statistic

```
. rdwinselect demmv dopen population demvoteslag1, stat(ksmirnov)
```

Window selection with smallest window including at least 10 observations in each group and adding 3 observations in each step

```
. rdwinselect demmv dopen population demvoteshlag1, obsmin(10) wobs(3)
```

Window selection setting smallest window at .5 and with .125 length increments

```
. rdwinselect demmv dopen population demvoteshlag1, wmin(.5) wstep(.125)
```

Window selection with large-sample p-values using 40 windows and a scatter plot

```
. rdwinselect demmv dopen population demvoteshlag1, nwindows(40) approximate plot
```

Modify graph options: add title and x-axis label

```
. rdwinselect demmv dopen population demvoteshlag1, nwindows(40) approx plot graph_options(title(Main title) xtitle(x-axis title))
```

Saved results

rdwinselect saves the following in **r()**:

Scalars

r(minp)	minimum p-value from covariate test
r(N)	sample size in recommended window
r(N_left)	sample size in recommended window to the left of the cutoff
r(N_right)	sample size in recommended window to the right of the cutoff
r(w_left)	left end of recommended window
r(w_right)	right end of recommended window
r(wobs)	when specified, increment (in observations) in each window
r(wmin_left)	left endpoint of initial window, when available
r(wmin_right)	right endpoint of initial window, when available
r(wstep)	when specified, increment (in window length) in each window
r(nwindows)	total number of windows evaluated

Locals

r(seed)	seed used in permutations
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Matrices

r(wlist_left)	matrix with left endpoints of candidate windows
r(wlist_right)	matrix with right endpoints of candidate windows
r(results)	matrix containing the minimum covariate-balance p-value, selected covariate index, binomial-test p-value, sample sizes, and window endpoints for each candidate window

References

- Cattaneo, M. D., Frandsen, B., and R. Titiunik. 2015. [Randomization Inference in the Regression Discontinuity Design: An Application to Party Advantages in the U.S. Senate.](#) *Journal of Causal Inference* 3(1): 1-24.
- Cattaneo, M.D., Titiunik, R. and G. Vazquez-Bare. 2016. [Inference in Regression Discontinuity Designs under Local Randomization.](#) *Stata Journal* 16(2): 331-367.
- Cattaneo, M. D., Titiunik, R. and G. Vazquez-Bare. 2017. [Comparing Inference Approaches for RD Designs: A Reexamination of the Effect of Head Start on Child Mortality.](#) *Journal of Policy Analysis and Management* 36(3): 643-681.

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